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**JOMO KENYATTA UNIVERSITY
OF
AGRICULTURE AND TECHNOLOGY**

University Examinations 2024/2025

**FIRST YEAR SECOND SEMESTER EXAMINATION FOR THE DEGREE OF BACHELOR OF
SCIENCE IN COMPUTER SCIENCE**

BIT 2111: COMPUTER AIDED DESIGN

DATE: MAY 2025

TIME: 2 HOURS

INSTRUCTIONS: Answer Question **ONE** and any other **TWO** Questions

QUESTION ONE

- (a) Define **geometric modeling** and explain its role in Computer-Aided Design (CAD) [4 Marks]
- (b) Differentiate between **rotation** and **movement** of objects in CAD, providing an example of each. [4 Marks]
- (c) Illustrate how to apply a hatch pattern in a CAD drawing. Describe a scenario where hatch patterns are useful. [6 Marks]
- (d) Explain how special effects, such as gradients and shadows, can be applied in a CAD environment to enhance the visual appearance of a drawing. [6 Marks]
- (e) What are **polylines**, and how do they differ from regular lines? [4 Marks]
- (f) Illustrate how you can draw a rectangle with dimensions **100 mm × 50 mm** and a **circle of radius 30 mm** inside it, centered at the midpoint. [6 Marks]

QUESTION TWO

- (a) Compare and contrast **absolute, relative, and polar coordinate systems** in CAD design. Provide real-world examples of when each system is most useful. [6 Marks]
- (b) Explain how you can rotate an **ellipse** by **45°** around its center and move it **30 mm to the right**. [5 Marks]
- (c) Explain how CAD has transformed industries such as **architecture, engineering, and product design**, providing real-world examples. [5 Marks]

- (d) Discuss the role of **text, labels, and special effects** in enhancing the readability and presentation of technical drawings. [4 Marks]

QUESTION THREE

- (a) Illustrate how to draw a line from **(10,10)** to **(50,10)** using absolute coordinates, then move it to **(20,30)** using relative coordinates. [6 Marks]
- (b) Explain the purpose of **dimensioning** in technical drawings. [4 Marks]
- (c) Identify current limitations in CAD tools and predict future trends, such as **cloud-based CAD, AI-driven design, and virtual reality integration**. [6 Marks]
- (d) Discuss the impact of **improper dimensioning** in CAD designs. Provide examples of how incorrect dimensions can affect manufacturing or construction projects. [4 Marks]

QUESTION FOUR

- (a) Evaluate the advantages and limitations of traditional hand drafting versus CAD-based design. [4 Marks]
- (b) You are working on a project that requires the modeling of a mechanical part. Discuss the **importance of modeling** in design and explain how different types of CAD models influence the final output. [6 Marks]
- (c) Explain how you can use the Cartesian coordinate system to draw a **triangle** in CAD by specifying absolute and relative coordinates. [4 Marks]
- (d) Transform a square object by **rotating** it 45 degrees and **scaling** it by 1.5 times. Document the coordinate changes before and after transformation. [6 Marks]